

CHAPTER 4

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4.1. If an ideal $\mathfrak{a}$ has a primary decomposition, then $\text{Spec}(A/\mathfrak{a})$ has only finitely many irreducible components.

Proof. Write $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ for its primary decomposition, with $\mathfrak{q}_i$ a $\mathfrak{p}_i$ -primary ideal for $i = 1, \dots, n$. We have that

$$\text{Spec}(A/\mathfrak{a}) \cong V(\mathfrak{a}) = \bigcup_{i=1}^n V(\mathfrak{q}_i) = \bigcup_{i=1}^n V(\mathfrak{p}_i).$$

Without loss of generality, take $m \leq n$ such that each $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ is minimal. Then, if $1 \leq i \leq m$, $V(\mathfrak{p}_i) \subset V(\mathfrak{p}) \subset V(\mathfrak{a})$ implies $\mathfrak{p}_i \supset \mathfrak{p} \supset \mathfrak{a}$, whence $\mathfrak{p}_i \supset \mathfrak{p} \supset \mathfrak{p}_j$ for $1 \leq j \leq m$. Minimality of $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ gives us that $i = j$, so $\mathfrak{p} = \mathfrak{p}_i$, showing that $V(\mathfrak{p}_i)$ is irreducible for all $1 \leq i \leq m$. We conclude that

$$V(\mathfrak{a}) = \bigcup_{i=1}^m V(\mathfrak{p}_i)$$

is a finite irreducible decomposition of $\text{Spec}(A/\mathfrak{a})$, as desired. $\square$

4.2. If $\mathfrak{a} = r(\mathfrak{a})$ is a decomposable ideal, then $\mathfrak{a}$ has no embedded prime ideals.

Proof. We can write

$$\mathfrak{a} = r(\mathfrak{a}) = \bigcap_{\mathfrak{p} \supset \mathfrak{a}} \mathfrak{p} = \bigcap_{i \in I} \mathfrak{p}_i,$$

where I is the (not necessarily finite) set of minimal elements in $\{\mathfrak{p} \supset \mathfrak{a}\}$. However, $\mathfrak{a}$ decomposable gives us that $I = \{1, \dots, n\}$ is finite, and in fact $\{\mathfrak{p}_i\}_{i=1}^n$ is the set of minimal ideals in the primary decomposition of $\mathfrak{a}$ by Proposition 4.6. We conclude that $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{p}_i$ is the intersection of its minimal ideals, whence it has no embedded ideals. $\square$

4.5. In the polynomial ring $k[x, y, z]$ with k a field, let $\mathfrak{p}_1 = (x, y)$, $\mathfrak{p}_2 = (x, z)$, $\mathfrak{m} = (x, y, z)$; $\mathfrak{p}_1$ and $\mathfrak{p}_2$ are prime, and $\mathfrak{m}$ is maximal. Let $\mathfrak{a} = \mathfrak{p}_1 \mathfrak{p}_2$. Show that $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$ is a reduced primary decomposition of $\mathfrak{a}$. Which components are isolated and which are embedded?

Proof. We have that $\mathfrak{a} = (x^2, xz, yx, yz) \subset \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$. All of the ideals are homogeneous, so it suffices to show the reverse inclusion for homogeneous elements. $y^2 \notin \mathfrak{p}_2$ and $z^2 \notin \mathfrak{p}_1$, so any homogeneous degree 2 element of $\mathfrak{m}^2$ is contained in $\mathfrak{a}$, so that the reverse inclusion holds.

$\mathfrak{m}$ maximal implies $\mathfrak{m}^2$ is $\mathfrak{m}$ -primary, and clearly each $\mathfrak{p}_i$ is $\mathfrak{p}_i$ -primary. $\mathfrak{m}^2$ contains no degree 1 elements, so $\mathfrak{p}_1, \mathfrak{p}_2 \not\subset \mathfrak{m}^2$. Conversely, $z^2 \in \mathfrak{m}^2 \setminus \mathfrak{p}_1$ and $y^2 \in \mathfrak{m}^2 \setminus \mathfrak{p}_2$ shows that neither of the reverse inclusions hold. Finally, $z \in \mathfrak{p}_2 \setminus \mathfrak{p}_1$ and $y \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$ show that neither contains the other. We conclude that our decomposition is reduced. Finally, $\mathfrak{p}_1, \mathfrak{p}_2 \supset \mathfrak{m}$ are minimal primes of $\mathfrak{a}$ by above, while $\mathfrak{m}$ is embedded. $\square$

4.12. Let A be a ring, S a multiplicatively closed subset of A . For any ideal $\mathfrak{a} \trianglelefteq A$, let $S(\mathfrak{a})$ denote the contraction of $S^{-1}\mathfrak{a}$ in A . $S(\mathfrak{a})$ is called the saturation of $\mathfrak{a}$ with respect to S . Prove that

- (a) $S(\mathfrak{a}) \cap S(\mathfrak{b}) = S(\mathfrak{a} \cap \mathfrak{b})$
- (b) $S(r(\mathfrak{a})) = r(S(\mathfrak{a}))$
- (c) $S(\mathfrak{a}) = (1) \iff \mathfrak{a}$ meets S
- (d) $S_1(S_2(\mathfrak{a})) = (S_1S_2)(\mathfrak{a})$.

If $\mathfrak{a}$ has a primary decomposition, prove that the set of ideals $S(\mathfrak{a})$ (where S runs over all multiplicatively closed subsets of A) is finite.

Proof. (a) We have that

$$\begin{aligned} S(\mathfrak{a} \cap \mathfrak{b}) &= (S^{-1}(\mathfrak{a} \cap \mathfrak{b}))^c \\ &= (S^{-1}\mathfrak{a} \cap S^{-1}\mathfrak{b})^c \\ &= (S^{-1}\mathfrak{a})^c \cap (S^{-1}\mathfrak{b})^c \\ &= S(\mathfrak{a}) \cap S(\mathfrak{b}). \end{aligned}$$

(b) We have that

$$\begin{aligned} S(r(\mathfrak{a})) &= (S^{-1}r(\mathfrak{a}))^c \\ &= (r(S^{-1}\mathfrak{a}))^c \\ &= r(S(\mathfrak{a})). \end{aligned}$$

(c) By 3.11ii, we have that

$$S(\mathfrak{a}) = \mathfrak{a}^{ec} = \bigcup_{s \in S} (\mathfrak{a} : s),$$

so $S(\mathfrak{a}) = (1)$ iff $1 \in (\mathfrak{a} : s)$ for some $s \in S$, i.e. $\mathfrak{a}$ meets S .

(d) We have

$$\begin{aligned} S_1(S_2(\mathfrak{a})) &= \bigcup_{s_1 \in S_1} (S_2(\mathfrak{a}) : s_1) \\ &= \bigcup_{s_1 \in S_1} \left(\bigcup_{s_2 \in S_2} (\mathfrak{a} : s_2) : s_1 \right) \\ &= \bigcup_{s_1 \in S_1} \{x \in A : xs_1 \in \bigcup_{s_2 \in S_2} (\mathfrak{a} : s_2)\} \\ &= \bigcup_{s_1 \in S_1} \{x \in A : \exists s_2 \in S_2 (xs_1s_2 \in \mathfrak{a})\} \\ &= \bigcup_{\substack{s_1 \in S_1 \\ s_2 \in S_2}} (\mathfrak{a} : s_1s_2) \\ &= (S_1S_2)(\mathfrak{a}). \end{aligned}$$

Finally, suppose $\mathfrak{a}$ has a primary decomposition $\mathfrak{a} = \bigcup_{i=1}^n \mathfrak{q}_i$, with each $\mathfrak{q}_i$ a $\mathfrak{p}_i$ -primary ideal. Fix a multiplicative subset $S \subset A$. We have that

$$\begin{aligned} S(\mathfrak{a}) &= S\left(\bigcap_{i=1}^n \mathfrak{q}_i\right) \\ &= \bigcap_{i=1}^n S(\mathfrak{q}_i), \end{aligned}$$

where

$$S(\mathfrak{q}_i) = \begin{cases} \mathfrak{q}_i & S \cap \mathfrak{p}_i = \emptyset \\ (1) & S \cap \mathfrak{p}_i \neq \emptyset \end{cases}$$

implies that

$$S(\mathfrak{a}) = \bigcap_{i \in I} \mathfrak{q}_i$$

for some $I \subset \{1, \dots, n\}$, possibly empty. There exist only finitely many I , concluding our proof. $\square$

4.13. Let A be a ring and $\mathfrak{p}$ a prime ideal of A . The n th symbolic power of $\mathfrak{p}$ is defined to be the ideal (with the notation of Exercise 4.12)

$$\mathfrak{p}^{(n)} := S_{\mathfrak{p}}(\mathfrak{p}^n),$$

where $S_{\mathfrak{p}} = A - \mathfrak{p}$. Show that

- (a) $\mathfrak{p}^{(n)}$ is a $\mathfrak{p}$ -primary ideal;
- (b) if $\mathfrak{p}^n$ has a primary decomposition, then $\mathfrak{p}^{(n)}$ is its $\mathfrak{p}$ -primary component;
- (c) $\mathfrak{p}^{(n)} = \mathfrak{p}^n \iff \mathfrak{p}^n$ is $\mathfrak{p}$ -primary.

Proof. (a) We have that

$$r(\mathfrak{p}^{(n)}) = S_{\mathfrak{p}}(r(\mathfrak{p}^n)) = S_{\mathfrak{p}}(\mathfrak{p}) = \mathfrak{p}.$$

(b) Write a minimal primary decomposition $\mathfrak{p}^n = \bigcap_{i=1}^n \mathfrak{q}_i$. Note that $\mathfrak{p} = r(\mathfrak{p}^n) = \bigcap_{i=1}^n \mathfrak{p}_i$ implies $\mathfrak{p}_i = \mathfrak{p}$ for some i . In fact, the above shows that $\mathfrak{p}$ is the unique minimal ideal in $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, so that $\mathfrak{p}_i \subset \mathfrak{p}$ implies $\mathfrak{p} = \mathfrak{p}_i$. Then, fixing $\mathfrak{p}_1 = \mathfrak{p}$,

$$\mathfrak{p}^{(n)} = \bigcap_{i=1}^n S_{\mathfrak{p}}(\mathfrak{q}_i) = \bigcap_{\mathfrak{p}_i \subset \mathfrak{p}_1} \mathfrak{q}_i = \mathfrak{q}_1$$

gives us our result.

(c) If $\mathfrak{p}^n$ is $\mathfrak{p}$ -primary, then it must equal its $\mathfrak{p}$ -primary component $\mathfrak{p}^{(n)}$. Conversely, $\mathfrak{p}^{(n)} = \mathfrak{p}^n$ implies $\mathfrak{p}^n$ is primary, since the former is primary. $\square$

4.15. Let $\mathfrak{a}$ be a decomposable ideal in a ring A , Σ an isolated set of prime ideals belonging to $\mathfrak{a}$, and let $\mathfrak{q}_{\Sigma}$ be the intersection of the corresponding primary elements. Let $f \in A$ such that, for each $\mathfrak{p} \supset \mathfrak{a}$ we have $f \in \mathfrak{p} \iff \mathfrak{p} \notin \Sigma$, and define $S_f = \langle f \rangle$ to be the powers of f . Show that $\mathfrak{q}_{\Sigma} = S_f(\mathfrak{a}) = (\mathfrak{a} : f^n)$ for all large n .

Proof. We have that $\Sigma \subset V(\mathfrak{a})$ is isolated iff $\Sigma = \bigcup_{\mathfrak{p} \in \Sigma} V(\mathfrak{p})$. Write a minimal primary decomposition $\mathfrak{a} = \bigcup_{i=1}^r \mathfrak{q}_i$, with each $\mathfrak{q}_i$ a $\mathfrak{p}_i$ -primary ideal. We have that $\mathfrak{p}_i \cap S_f = \emptyset \iff f \notin \mathfrak{p}_i \iff \mathfrak{p} \in \Sigma$, so that $\mathfrak{q}_\Sigma = S_f(\mathfrak{a})$ follows from Proposition 4.9. Note that

$$S_f(\mathfrak{a}) = \bigcap_{i=1}^r S_f(\mathfrak{q}_i) = \bigcap_{i=1}^r \bigcup_{n \geq 1} (\mathfrak{q}_i : f^n).$$

$xf^n \in \mathfrak{q}_i$ implies either $x \in \mathfrak{q}_i$ or $f^{m_i} \in \mathfrak{q}_i$ for some $m_i \geq 1$. If $f^{m_i} \in \mathfrak{q}_i$, then

$$\bigcup_{n \geq 1} (\mathfrak{q}_i : f^n) = (\mathfrak{q}_i : f^{m_i}) = (1).$$

Otherwise, $\bigcup_{n \geq 1} (\mathfrak{q}_i : f^n) = (\mathfrak{q} : f^m) = \mathfrak{q}$ for any arbitrary $m \geq 1$. Then, for any $m \geq \max\{m_i\}$, we have

$$S_f(\mathfrak{a}) = \bigcap_{i=1}^r (\mathfrak{q}_i : f^m) = (\mathfrak{a} : f^m).$$

□

4.18. Consider the following conditions on a ring A :

- (L1) For every proper ideal $\mathfrak{a} \subsetneq A$ and every $\mathfrak{p} \in \text{Spec } A$, there exists $x \notin \mathfrak{p}$ such that $S_{\mathfrak{p}}(\mathfrak{a}) = (\mathfrak{a} : x)$, where $S_{\mathfrak{p}} = A \setminus \mathfrak{p}$.
- (L2) Given an ideal $\mathfrak{a}$ and a descending chain $S_1 \supset S_2 \supset \dots$ of multiplicatively closed subsets of A , there exists an integer n such that $S_n(\mathfrak{a}) = S_{n+1}(\mathfrak{a}) = \dots$.

Prove that the following are equivalent:

- (a) Every ideal in A has a primary decomposition;
- (b) A satisfies (L1) and (L2).

Proof. (a $\implies$ b) We first prove (L1). Let $\mathfrak{a} \subsetneq A$ and take $\mathfrak{p} \in \text{Spec } A$. If $\mathfrak{a} \not\subset \mathfrak{p}$, then we can take $x \in \mathfrak{a} \setminus \mathfrak{p}$ so that $S_{\mathfrak{p}}(\mathfrak{a}) = (1) = (\mathfrak{a} : x)$.

Now, suppose we have $\mathfrak{a} \subset \mathfrak{p}$, and consider the isolated set of prime ideals

$$\Sigma = V(\mathfrak{p}) \subset V(\mathfrak{a})$$

over $\mathfrak{a}$.

□